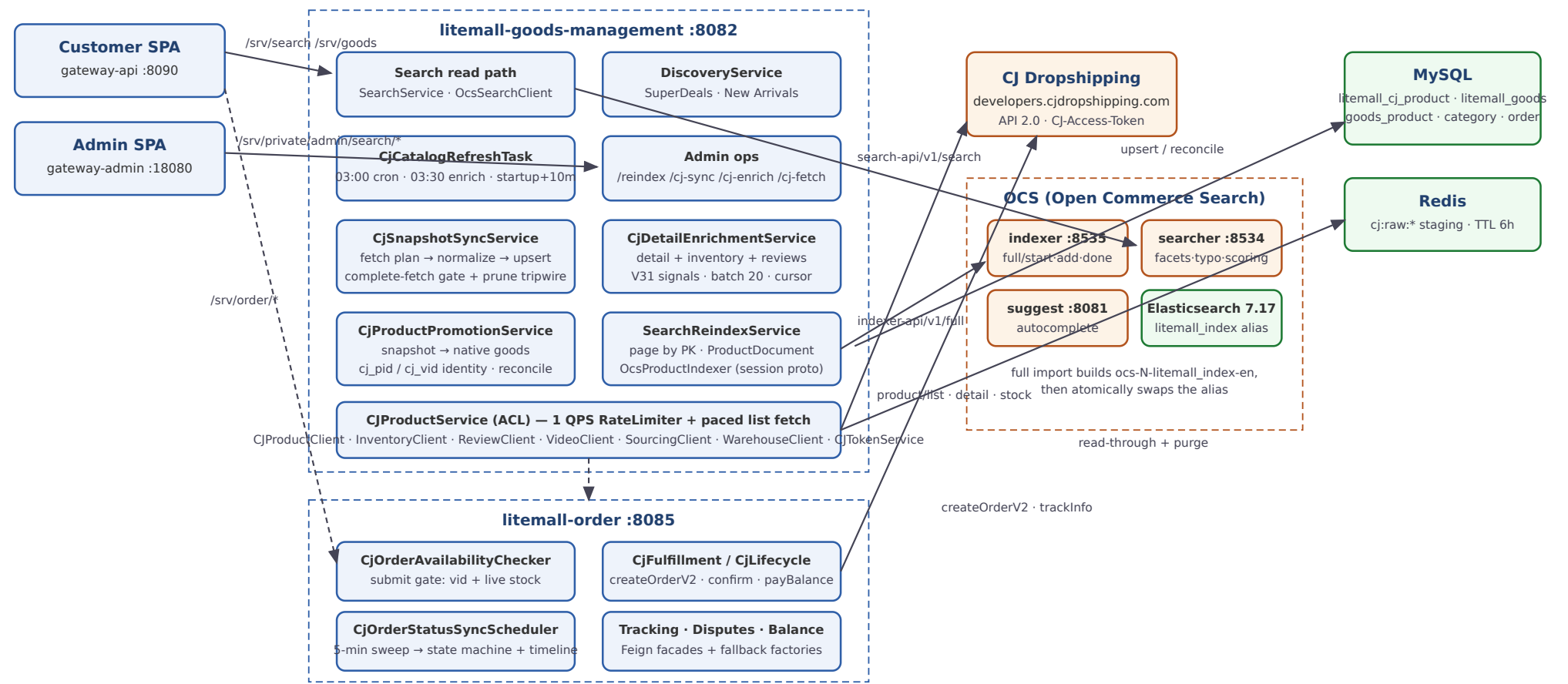


CJ Dropshipping Pipeline — Architecture & Sequence Reference

litemall microservices · 2026-07-13 · covers CJ fetching, catalog sync, OCS indexing, search/ranking, and order-side fulfillment · includes the catalog-erosion incident fix (master db0c4eaaaf)

Two modules integrate CJ independently: **litemall-goods-management** (catalog fetch/sync/enrich/promote + OCS indexing + search) with RestTemplate clients, and **litemall-order** (stock gate, createOrderV2 lifecycle, tracking, disputes) with Feign clients. Each has its own token cache. All CJ calls carry the `CJ-Access-Token` header and respect the ~1 QPS quota.

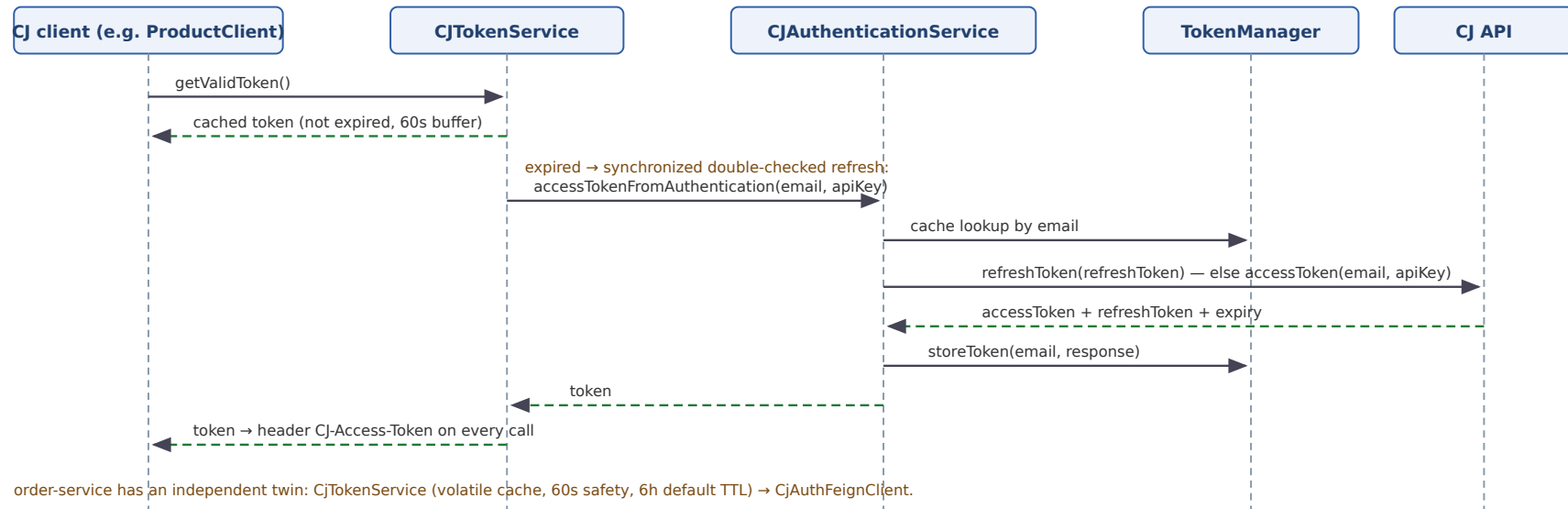
1 · System architecture



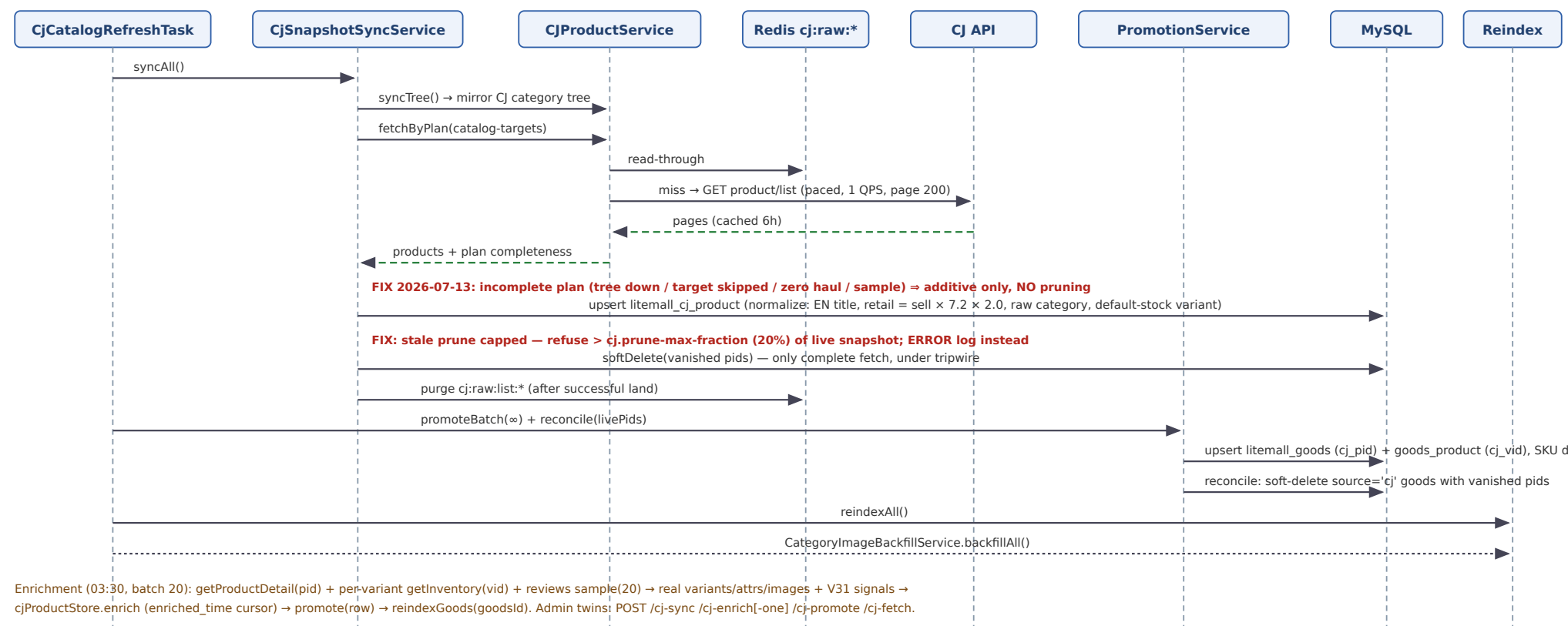
2 · Component inventory

Component	Module	Responsibility	External deps
CJTokenService · TokenManager · CJAuthenticationService	goods-mgmt	Token acquire/cache (60s expiry buffer), attach CJ-Access-Token	CJ auth API
CJRequestUtils + clients (Product / Inventory / Review / Video / Sourcing / Warehouse)	goods-mgmt	RestTemplate calls with token header	CJ REST
CJProductService	goods-mgmt	Rate-limited (1 QPS + paced list) Redis read-through facade: list, detail, inventory, reviews, videos, sourcing, warehouse	CJ REST, Redis
CjRawCacheRepository	goods-mgmt	Raw payload staging under cj:raw:*, TTL 6h, list purge after successful sync	Redis
CjCatalogRefreshTask	goods-mgmt	Schedules: refresh 03:00, enrich 03:30, one-shot startup refresh (+10 min)	—
CjSnapshotSyncService	goods-mgmt	Plan-driven fetch → normalize (English title, retail = sell × usdToCny × margin, raw category path) → upsert snapshot; stale prune only on complete fetch, capped by tripwire (2026-07-13 fix)	CJ API, MySQL, Redis
CjDetailEnrichmentService	goods-mgmt	Detail + per-variant inventory + review aggregate → V31 signals; batch 20, enriched_time cursor; promote + single-doc reindex	CJ API, MySQL
CjProductPromotionService · CjProductToNativeAdapter	goods-mgmt	Snapshot → native goods: pid-litemall_goods.cj_pid, vid-goods_product.cj_vid, SKU diff, ranking-signal update, category resolve, reconcile() soft-delete of vanished pids	MySQL
CjCategoryTreeSyncService	goods-mgmt	Mirror the 3-level CJ tree into litemall_category; leaf-by-path resolution	CJ API, MySQL
SearchReindexService · LitemallProductIndexingService · OcsProductIndexer	goods-mgmt	Page goods by PK → ProductDocument (V31 signals emitted 0-not-absent, facet attrs, per-SKU variants) → OCS full-import session / single upsert	OCS indexer :8535
SearchService · OcsSearchClient · OcsSuggestClient · DiscoveryService	goods-mgmt	Query/facets/filters/sort/typo → ranked unified list (source tag local cj); SuperDeals (–listed_num), New Arrivals (–created_epoch + price band)	OCS searcher :8534, suggest :8081
LitemallGoodsController#listGoods	goods-mgmt	ocsServable flag: OCS path with DB fallback when OCS is down / brand-isNew-isHot queries	OCS, MySQL
CjGoodsVideoService · comments proxy	goods-mgmt	CJ videos (ON_STATE only) + review proxy for detail pages, Redis 6h	CJ API, Redis
CjTokenService (order)	order	Independent volatile token cache (60s safety, 6h default TTL)	CJ auth API
CjOrderAvailabilityChecker	order	Pre-TX submit gate: cj_vid present + live stock ≥ qty (advisory — CJ down passes)	CJ stock API
CjFulfillmentService · CjLifecycleService · CjDropshipOrderFacadeImpl	order	Pay-time createOrderV2 (payType=3 draft, sandbox, iossType) → confirm → optional payBalance; status mapping into the state machine + timeline	CJ REST (Feign)
CjOrderStatusSyncScheduler	order	5-min sweep of open CJ orders (batch 20, 1.1s pace)	—
CjTrackingFacadeImpl · CjDisputeService · balance	order	trackInfo (1h cache), dispute vertical, account balance	CJ REST (Feign)

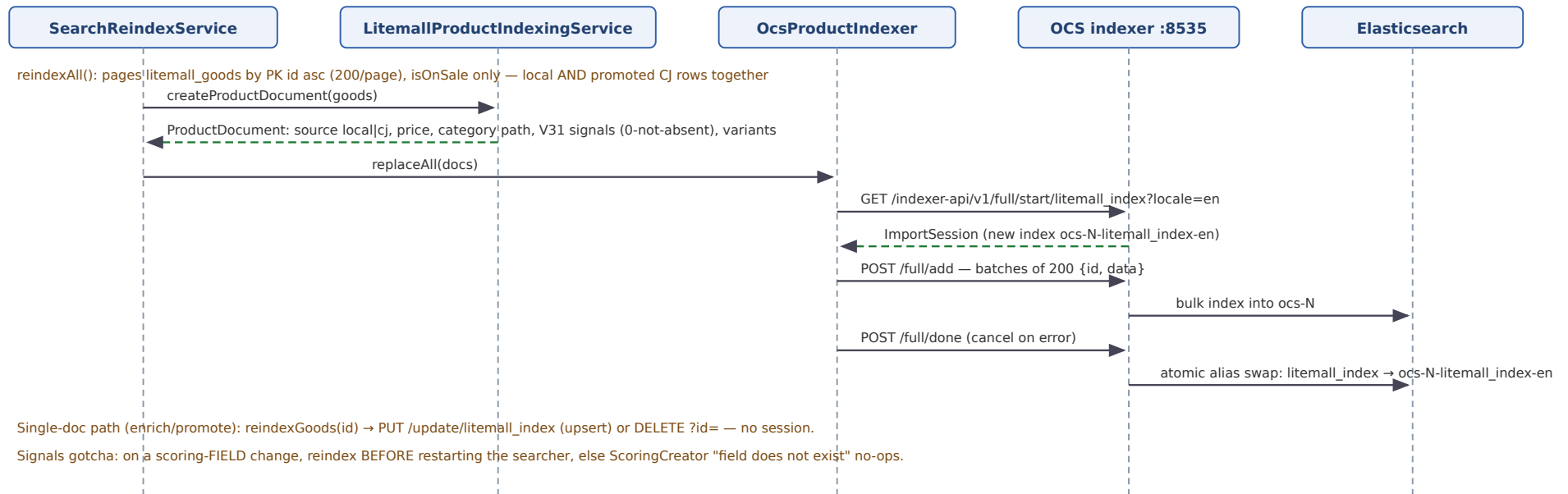
3 · Sequence — CJ auth token



4 · Sequence — nightly catalog refresh & sync (03:00 + startup one-shot)



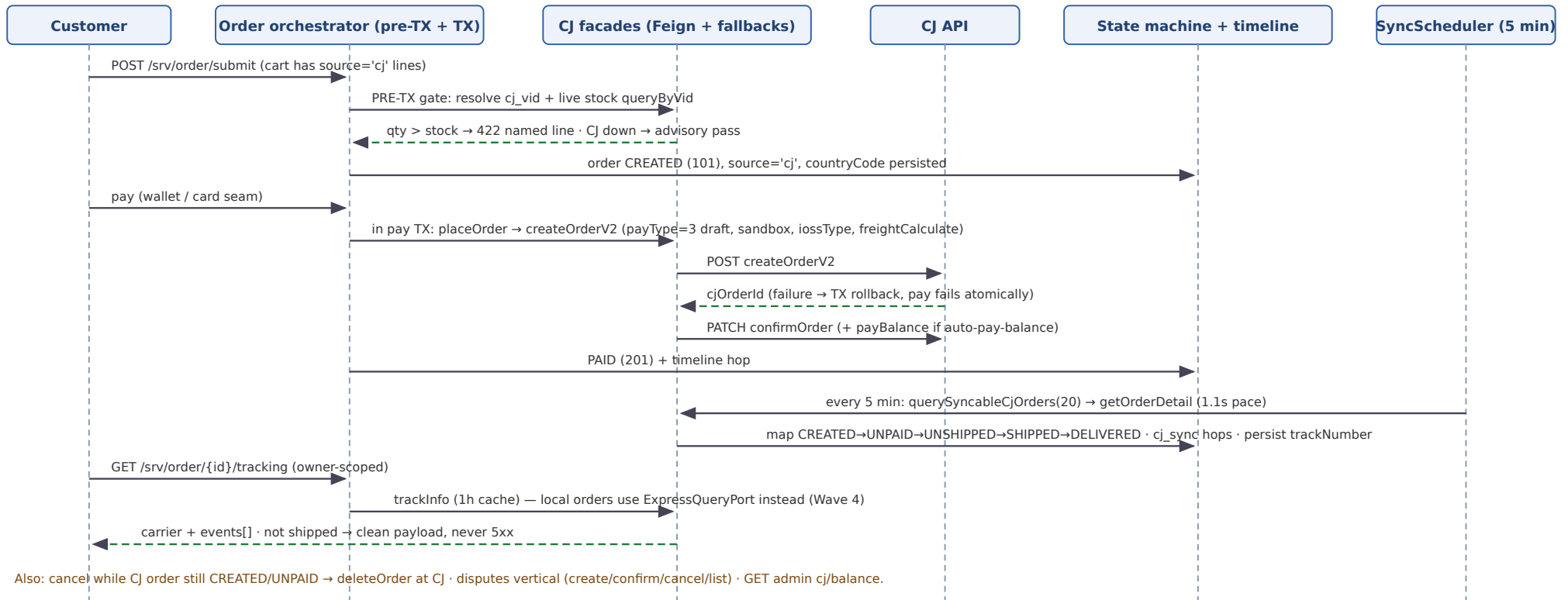
5 · Sequence — OCS indexing (full import session)



6 · Sequence — customer search & ranking



7 · Sequence — order-side CJ lifecycle



8 · Ranking model (V31 signals)

Signal	Index field	Scoring	Source (local)	Source (CJ)
In-stock	stock (variant-level)	ln1p multiply — out-of-stock scores 0 (hard demote)	goods_product number	enriched per-variant CJ inventory (default 100 pre-enrich)
Popularity	listed_num	ln2p multiply (ln(2+x): no signal ⇒ uniform ~0.69, never zeroes)	sales counter	CJ listedNum (free via enrichment)
Reviews	review_count, rating	ln2p multiply each	litemall_comment backfill (refresh-signals)	CJ review aggregate (sample 20)
Recency	created_epoch	sort field for New-Arrivals (–created_epoch); no decay function	add_time	cjCreateTime

All factors are `field_value_factor` with `boost-mode: multiply`, `score-mode: multiply` — text relevance × in-stock × popularity × reviews. Discovery rails reuse the same index: **SuperDeals** = sort -listed_num; **New Arrivals** = sort -created_epoch + price band 0–affordable-max-price (100). Query ladder: AND-fuzzy → relaxed → ngram(80%) → fallback, with Querqy synonym expansion; weighted fields title 4 / category 2.5 / brand 2 / description 1.

9 · Schedules & config keys (verbatim)

What	Key	Default
Nightly catalog refresh	spring.cjdropship.refresh-cron	0 0 3 * * *
Nightly enrichment	spring.cjdropship.enrich-cron	0 30 3 * * * (batch enrich-batch-size: 20)
Startup one-shot refresh	refresh-on-startup / refresh-startup-delay-ms	true / 600000 (10 min)
Erosion tripwire (new)	spring.cjdropship.prune-max-fraction	0.2 — raise deliberately for an intentional rebuild
List-fetch pacing / page size	fetch-pace-seconds / page-size	300 / 200
Raw-payload staging	redis.raw-ttl-seconds / redis.purge-after-sync	21600 (6 h) / true
Retail pricing	pricing.usd-to-cny / pricing.margin	7.2 / 2.0
CJ credentials / endpoint	spring.cjdropship.api.{auth.cj-email, auth.cj-api-key, base-url}	env-supplied
Order-side lifecycle	litemall.order.cj-sync-sweep-ms / cj-sync-batch; api.{auto-pay-balance, sandbox, ioss-type, from-country-code, logistic-name}	300000 / 20; true / false / 1 / CN / CJPacket Ordinary
Search endpoints	litemall.search.{indexer-url, search-url, suggest-url, index-name, locale}	:8535 / :8534 / :8081 / litemall_index / en
Discovery band	litemall.discovery.affordable-max-price	100

10 · Incident appendix — catalog erosion (fixed 2026-07-13)

Symptom: CJ products no longer fetched or ranked; search nearly empty. **Found:** 7,779 of 7,783 CJ goods (and 7,781 snapshot rows, 13,362 SKUs) soft-deleted in waves (07-05: 582, 07-07: 3,366, 07-10: 3,710) — each a nightly/startup refresh.

Root cause: `CjSnapshotSyncService` stale-pruning treated "absent from this fetch" as "vanished upstream". The guard only covered a fully-EMPTY fetch, but the plan-driven fetch is routinely PARTIAL: a failed category-tree call silently skips all name-based targets, a target can resolve to nothing or return zero, and CJ's rotated per-category listings change which products a limited haul sees. Every unseen product got soft-deleted, and `CjProductPromotionService.reconcile()` cascaded the deletions into the native goods.

Fix (master db0c4eaaf): (1) the fetch plan now reports *completeness* — an incomplete full sync is additive-only (no pruning, WARN); (2) an erosion tripwire (`prune-max-fraction` , default 20%) refuses any prune above that fraction of the live snapshot with an ERROR log. **Recovery:** catalog restored by un-deleting the eroded rows; signals refreshed; full reindex rebuilt `literal_index` with 8,021 docs (7,783 CJ + local); verified: search totals, source=cj filter, popularity/new-arrival sorts, SuperDeals rail.